

JABATAN TEKNOLOGI MAKLUMAT DAN KOMUNIKASI

**DIPLOMA TEKNOLOGI MAKLUMAT
(TEKNOLOGI DIGITAL)**

**FINAL YEAR PROJECT
PROPOSAL REPORT**

Omni-View Business Command Centre

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1.0 INTRODUCTION

Capt Empire is a retail business specializing in the sale of shirts through both physical and online sales channels. As the business grows, managing daily operations such as sales records, inventory, employee performance, and Key Performance Indicators (KPIs) becomes increasingly important. Relying on manual record-keeping or multiple disconnected tools can lead to inaccurate data, inefficient workflows, and difficulties in monitoring overall business performance.

To address these challenges, this project proposes the development of a web-based Integrated Operations Management System for Capt Empire. The system is designed to provide a centralized platform for managing the company's internal operations by integrating multiple business functions into a single application. This enables management to efficiently monitor sales performance, inventory levels, employee productivity, and operational KPIs.

The proposed system will include modules for sales management, inventory management, employee management, KPI tracking, and reporting. An administrative dashboard will present key business information through charts and summaries, allowing management to analyze sales trends, evaluate employee performance, monitor stock levels, and make informed business decisions.

By digitalizing and integrating operational processes, the proposed system is expected to improve data accuracy, reduce manual workload, enhance operational efficiency, and support data-driven decision-making. Ultimately, the system aims to strengthen Capt Empire's business management by providing a reliable and user-friendly platform for its daily operations.

2.0 PROBLEM STATEMENT

Capt Empire currently relies on manual processes and separate records to manage its daily business operations. Sales transactions, inventory updates, and employee performance data are not maintained within a centralized system, making it difficult to access accurate and up-to-date information. This fragmented approach increases the risk of data inconsistencies, duplicate records, and human errors.

Furthermore, the company lacks an effective mechanism to monitor employee performance and Key Performance Indicators (KPIs). Without a centralized dashboard, management faces difficulties in evaluating staff productivity, measuring business performance, and identifying areas that require improvement. Preparing reports and analyzing sales trends also require significant manual effort, reducing operational efficiency and delaying decision-making.

In addition, inventory monitoring is not integrated with sales records, making it challenging to maintain accurate stock information and identify fast-moving or low-stock products. This limits the company's ability to plan inventory effectively and optimize business operations.

Therefore, there is a need for a web-based Integrated Operations Management System that centralizes sales management, inventory control, employee performance monitoring, KPI tracking, and business reporting. Such a system will improve operational efficiency,

enhance data accuracy, and provide management with valuable insights for informed decision-making.

3.0 OBJECTIVE

1. To design and develop a web-based Integrated Operations Management System:

System for Capt Empire that centralizes sales, inventory, employee, and KPI management.

2. To implement sales and inventory management modules:

Enable efficient recording, monitoring, and analysis of business transactions and stock levels.

3. To develop an employee performance and KPI monitoring module:

Assists management in evaluating staff productivity and operational performance.

4. To implement an administrative dashboard :

Provide a dashboard with reports and data visualizations that support business analysis and informed decision-making.

4.0 SCOPE

Admin Dashboard Features:

1. Product management (Create, Read, Update, Delete operations)
2. Stock inventory management with easy-to-use interface for status updates
3. Sales records tracking (transaction date, product, quantity, revenue)
4. Staff performance dashboard displaying basic metrics (sales volume, transaction count)
5. Admin user authentication and access control
6. Staff are able to interact with an AI-powered chatbot to obtain answers to sales-related questions.
7. Database backend with relational data structure

Technical Implementation:

1. Full-stack web application (frontend + backend)
2. Relational database for data persistence
3. Server-side business logic and data validation

Customer-Facing Features:

1. Product catalog with display of item details (name, description, images, price)
2. Near-real-time stock status display (Available, Limited Stock, Sold Out)
3. Size reference chart and/or sizing calculator tool
4. Direct redirect links to Shopee and TikTok Shop for customer checkout
5. Responsive web interface for browsing

5.0 PROJECT SIGNIFICANT

This system directly addresses operational bottlenecks that impact customer satisfaction and business efficiency. By centralizing stock visibility and reducing customer inquiries through live chat, the company can allocate support resources more effectively. The integrated analytics dashboard provides actionable business intelligence for informed decision-making, enabling management to track sales trends and staff performance in real time.

Customer Experience Improvement:

Customers benefit from immediate, self-service access to product availability and accurate sizing information, reducing friction in the shopping journey and decreasing return rates due to sizing errors. This improves customer satisfaction and builds trust in the brand.

Operational Efficiency:

Staff can manage inventory updates seamlessly through an intuitive admin interface. Management gains visibility into staff performance metrics, enabling data-driven performance evaluation and resource allocation.

Technical Relevance:

The project demonstrates full-stack web development proficiency, integrating frontend technologies (HTML, CSS, JavaScript, Bootstrap) with modern backend infrastructure (Supabase/PostgreSQL). It showcases practical experience in database design, authentication systems, and real-time data management — skills critical in contemporary e-commerce development.

Scalability & Future Growth:

The modular architecture allows for future enhancements, such as advanced analytics, inventory forecasting, or API integrations with e-commerce platforms, positioning the company for sustainable growth.

6.0 LITERATURE REVIEW

Modern small retail businesses increasingly turn to web-based platforms to manage inventory, sales, and customer interaction in one place. **Shopify** is one of the most widely adopted examples, offering real-time inventory tracking across sales channels and locations, with stock automatically syncing whenever a sale occurs (Shopify, 2026). Its dashboard also generates analytics like sell-through rate and days-of-stock-remaining. However, Shopify is a general-purpose commerce platform, meaning small clothing sellers pay for far more than they need, with no staff performance tracking, and social commerce support limited mainly to Instagram/Facebook rather than TikTok.

NetSuite Inventory Management is a more enterprise-focused example built with apparel businesses in mind, offering KPI dashboards and demand-based planning across warehouses and multiple channels (NetSuite, 2025). It's strong for scale but requires costly licensing and IT overhead unreasonable for a small single-location business, and still lacks any chatbot or staff KPI feature.

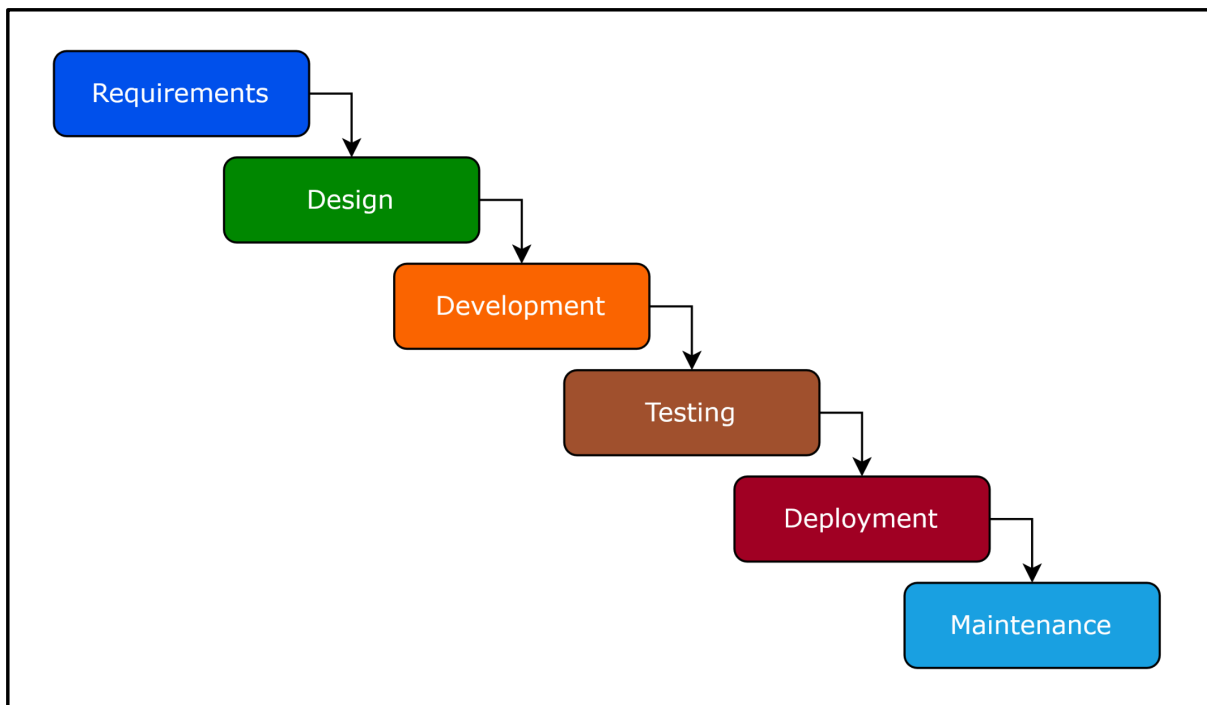
Square for Retail is the closest match in target scale, functioning as a complete inventory system for small businesses with real-time stock tracking and reporting synced across channels (Square, 2026a), plus dashboard-based stock adjustment (Square, 2026b). Still, it's a paid third-party SaaS with transaction fees, no staff KPI tracking, no AI assistant, and no TikTok support.

TikTok Shop's product detail pages serve as the benchmark for our customer-facing module, providing customers with product details such as name, description, price, images, size, and shipping options to support informed purchasing decisions (TikTok Shop Seller University, 2026).

Together, these systems show that inventory tracking and sales dashboards are already common in commercial retail platforms, but none combine inventory management, sales analytics and staff KPI tracking, a low-cost system built for small clothing businesses selling via TikTok live.

7.0 METHODOLOGY

This project follows a Waterfall methodology with incremental demonstrations. The development lifecycle is divided into distinct phases: requirements, design, development, testing, deployment and maintenance. Each phase completes before the next begins, with three demonstrations (Week 6, 9, 12) to validate progress and gather feedback.



Waterfall Diagram 7.1

Phase 1 is Project Proposal and Planning (Week 1 – Week 4). The project scope, objectives, and system requirements are defined. Client consultations are conducted to confirm staff performance metrics and analytics requirements. The project plan and system design are prepared, while the development environment is configured using Visual Studio Code, Git, and GitHub. The Supabase project is initialized, including database configuration and authentication setup. The project proposal is presented in Week 4 (10%).

Phase 2 is System Design and Configuration (Week 5 – Week 6). The Supabase PostgreSQL database tables and relationships for products, stock, staff, sales, and administrator accounts are designed and configured. Low-fidelity mock-ups of the customer interface and admin dashboard are created. The user interface is designed using HTML5, CSS3, Bootstrap, and JavaScript. The system architecture and data flow between the frontend and Supabase backend are planned, and frontend development for customer-facing features begins. Demonstration 1 is conducted in Week 6, showcasing the product catalogue, stock status, and sizing reference tool (10%).

Phase 3 is Development (Week 7 – Week 9). The admin dashboard is developed to support product management, stock updates, sales tracking, staff performance monitoring, and KPI reporting. Redirect links to Shopee and TikTok Shop are implemented. An AI-powered sales assistant chatbot is integrated into the admin dashboard using an AI provider API, enabling administrators to query sales-related topics and receive intelligent, context-aware responses. The frontend is connected to the Supabase backend via the Supabase JavaScript SDK, and data integration is tested to ensure all system components function correctly. Demonstration 2 is conducted in Week 9 to present the admin dashboard and additional features (10%).

Phase 4 is Testing (Week 10 – Week 12). Manual functional testing is performed on all system features, including the customer interface, administrator operations, and database functionality. Integration testing is conducted between the frontend and Supabase backend. User Acceptance Testing (UAT) is carried out with client feedback where possible, followed by bug fixing and user interface improvements. Demonstration 3 (15%) and the Final Presentation (30%) are conducted in Week 12.

Phase 5 is Deployment (Week 13). Following successful testing and refinement, the system is deployed to a hosting environment to make it accessible for actual use. This involves final configuration of the production environment, verifying that the Supabase backend connections remain functional post-deployment, and confirming that all core features, including the product catalogue, admin dashboard, and chatbot integration, operate correctly outside the local development setup.

Phase 6 is Maintenance (Week 14 - Week 15). The technical report is prepared to document the system architecture, database design, implementation, and testing process. The project logbook is completed by recording weekly progress, challenges, and solutions, while a user guide is prepared for administrator dashboard operations. This phase also outlines the planned approach to post-handover maintenance, including how bug fixes and minor client-requested changes will be handled after project completion. The Technical Report (15%) and Logbook (10%) are submitted in Week 15.

8.0 REFERENCES

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9.0 GANTT CHART

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10.0 TOOLS, MATERIALS & COST

Item	Quantity	Unit Cost (per year)	Total Cost (per year)
Software & Tools	-	Free	RM0
Hosting	-	RM 180	RM180
Domain	-	RM 45	RM45
AI API Key	-	Free tier	RM0
Total:			RM225

11.0 CONCLUSION

In conclusion, the proposed **“Omni-View Business Command Centre”** aims to improve the company's internal business operations by providing a centralized platform for managing sales records, inventory, employee performance, and Key Performance Indicators (KPIs). By replacing manual processes with a digital system, the project is expected to improve data accuracy, reduce administrative workload, and increase operational efficiency.

The system will also provide management with valuable insights through dashboards and reports, enabling better monitoring of business performance and supporting data-driven decision-making. These features will help the company identify sales trends, evaluate employee productivity, monitor inventory levels, and improve overall business management.

The significance of this project extends beyond meeting the current operational needs of the company. It demonstrates the practical application of information technology in solving real-world business problems while providing a scalable foundation for future enhancements such as advanced analytics, automated reporting, and integration with other business systems.

This project serves as an opportunity to apply the knowledge and technical skills acquired throughout the Diploma in Information Technology programme in developing a practical and beneficial solution for a real business environment.